

Practice Tests 2

Course Name Database Principle Examination Time 120minutes

Student Number _____ Student Name _____

1. Answer the following questions: (4+4+4+4+4)
 - (1) In database design, is the higher the NF(Normal Form) that the data schema follows, the better?
 - (2) Explain the ACID properties of “Transaction”?
 - (3) List those sub-constraints included in Static constraints. Which sub-constrain contains Domain constraints? And explain what’s Domain constraints.
 - (4) List three recovery strategies in DBMS, and describe the necessity of the commit rule and the log ahead rule for updating transactions.
 - (5) Does dense index always improve the efficiency of queries on the indexed attribute? Why?
2. Try to answer the following questions about concurrency in database systems.
 - (1) What is concurrency?(1)
 - (2) Why concurrency?(1)
 - (3) What problems will concurrency bring?(2)
 - (4) What kind of concurrent execution is correct?Why?(3)
 - (5) How to avoid problems caused by concurrency?(3)
3. What is the basic idea of the locking protocol and how does it ensure that concurrent transactions are executed correctly? What are the problems that must be solved after the adoption of the locking protocol? (5+5)
4. Consider a relational table:
Professor(
 professor_name, professor_id,
 professor_office_id, student_id,
 student_name, student_office_id,
 student_designated_refrigerator_id, refrigerator_owner_id,
 refrigerator_id, refrigerator_size, secretary_name,
 secretary_id, secretary_office)
Suppose the data has the following properties:
 - A. Professors and secretaries have individual offices, students share offices.
 - B. Students can work for multiple professors.
 - C. Refrigerators are owned by one professor.
 - D. Professors can own multiple refrigerators.
 - E. Students can only use one refrigerator.
 - F. The refrigerator the student uses must be owned by one of the professors they

work for.

G. Secretaries can work for multiple professors.

H. Professors only have a single secretary (

- (1) Put this table into 3rd normal form by writing out the decomposed tables; designate keys in your tables by underlining them. Designate foreign keys by drawing an arrow from a foreign key to the primary key it refers to. Note that some of the properties listed above may not be enforced (i.e., guaranteed to be true) by a 3NF decomposition.(5)
- (2) Which of the eight properties (A–H) of the data are enforced (i.e., guaranteed to be true) by the 3NF decomposition and primary and foreign keys you gave above.(5)
- (3) Please draw the E-R diagram of the relationship in this question.(10)

5. Hotel Management System introduction:

One room have one or several beds, and each room has information of room ID (e.g., 201), price and bed number.

Each customer has information of ID, name and gender.

Each reserve record has information of the ID of customer, the ID of room and reserve time.

Each room can be reserved by only one customer at the same time.

Table:

Room(RID, Price, Number);

Customer(CID, Name, Gender);

Reserve(RID, CID, Rdate)

- (1) For the relationships in the this question, we have the following query(10):

```
SELECT Name
```

```
FROM Room, Customer, Reserve
```

```
WHERE Room.RID = Reserve.RID AND
```

```
Customer.CID = Reserve.CID AND
```

```
Room.Price > 500 AND
```

```
Customer.Gender = 'female' AND
```

```
Reserve.Rdate between [2019.10.01, 2019.10.07]
```

Please optimize the query and draw the query trees before and after optimization.

- (2) For the Room table in this question, apply Assertion to ensure that the number of rooms ≥ 0 . (5)

6. For the three relational schemas in the above application scenario, write SQL statements that express the following query requirements (must be expressed in a single SQL statement, and WITH is only accepted in question):

- (1) Use join query to find the name of the guest who booked room 210;(5)
- (2) Query the name of the guest who booked all rooms;(5)
- (3) Query the room number and guest name booked by only one person in January 2016;(10)

- (4) For guests who have booked a room more than 60 times, query the maximum number of room bookings per guest in a single month and the year and month corresponding to the maximum number.(10)

Note: The functions for obtaining the year and month of Datetime data in SQL are getYear(...) and getMonth(...) respectively.

7. The compatibility matrix of (S, X) locks is shown in Figure (a). Suppose that we change the compatibility matrix of (S, X) locks to the one as shown in Figure (b). Now we have n transactions $\{T_1, T_2, \dots, T_n\}$ that require concurrent access in a period of time. Assume that these n transactions are read operations. Will the system work properly? If yes, explain the reason; If not, what happens? (10)

	Locks already owned by other transactions		
Lock request		S	X
	S	Y	N
	X	N	N

(a) Compatibility Matrix A

	Locks already owned by other transactions		
Lock request		S	X
	S	Y	Y
	X	N	N

(b) Compatibility Matrix B

8. Please answer the following questions: (5+5+10=20)

- (1) Please explain what is Trigger and list the components of Trigger.
- (2) Design a trigger for the SelectCourse table (in the fourth question) to ensure that $0 \leq \text{Grade} \leq 100$ and Sno exists in Student table and Cno exists in Course table when a tuple is inserted.
- (3) Try to define a trigger on the accommodation relationship table in question 2 to achieve the following function: For each Insert statement, if the inserted record does not have a check-out date, the operation will be rolled back.

Answer

1. Answer the following questions:

- (1) In database design, is the higher the NF(Normal Form) that the data schema follows, the better?
- (2) Explain the ACID properties of “Transaction”?
- (3) List those sub-constraints included in Static constraints. Which sub-constrain contains Domain constraints? And explain what’s Domain constraints.
- (4) List three recovery strategies in DBMS, and describe the necessity of the commit rule and the log ahead rule for updating transactions.
- (5) Does dense index always improve the efficiency of queries on the indexed attribute? Why?

(1)

In database design, it is not always better for the data schema to follow a higher Normal Form (NF). The choice should depend on the specific application. The primary purpose of database normalization is to prevent data redundancy and to avoid update, insertion, and deletion anomalies. However, a higher level of normalization can lead to slower processing speeds and more complex processing logic, which can decrease database performance. Therefore, a trade-off is necessary.

数据库设计中，数据模式遵循的范式不是越高越好，应取决于应用。数据库的范式主要目的是防止数据冗余，更新异常、插入异常和删除异常，而范式高会存在处理速度缓慢和处理逻辑复杂的问题，从而降低数据库性能，因此需要权衡考虑。

(2)

A transaction T is a finite sequence of actions on DB exhibiting the following effects:

Atomic action: Nothing or All.

Consistency preservation: consistency state of DB $\rightarrow$ another consistency state of DB.

Isolation: concurrent transactions should run as if they are independent each other.

Durability: The effects of a successfully completed transaction are permanently reflected in DB and recoverable even failure occurs later.

事务 T 是 DB 上的有限动作序列，表现出以下效果：

A 原子动作：无或全部。

C 一致性保持：DB 的一致性状态 $\rightarrow$ DB 的另一种一致性状态

I 隔离：并发事务应该像彼此独立一样运行。

D 持久性：成功完成的事务的效果会永久地反映在 DB 中，即使稍后发生故障也可以恢复。

(3)

Static constraints contain:

inherent constraints

implicit constraints

explicit constraints

implicit constraints contain Domain constraint.

Domain constraints: Field values must be of right type. Always enforced.

(4)

Three recovery strategies in DBMS:

A.I $\rightarrow$ DB before commit,

A.I $\rightarrow$ DB after commit,

A.I $\rightarrow$ DB concurrently with commit

摘自《数据库系统教程（王能斌）》

1. 后像在事务提交前完全写入数据库
2. 后像在事务提交后才写入数据库
3. 后像在事务提交前后写入数据库

The commit rule is to ensure that the A.I(after image) is written into the non-volatile memory before the transaction is committed, so that even if a failure occurs after the transaction enters the commit stage, the recorded A.I can still be used to redo and update, so as to ensure that the transaction meets the ACID principle.

The log ahead rule means that if the A.I is directly written to the database before the transaction is committed, the corresponding B.I(before image) must be written to the log before the transaction is committed, so that undo can be done when a failure occurs before the transaction enters the commit stage, and the execution of the transaction meets the ACID principle.

要点:提交规则是保证后像在事务提交前写入非易失存储器,这样即使事务进入提交阶段以后发生故障,仍可利用记录下来的后像重做更新,从而确保事务满足 ACID 原则;

先记后写规则是指如果后像在事务提交前直接写入数据库,则必须在此之前将相应的前像写入运行记录(Log),以备当事务进入提交阶段之前发生故障时做 undo,使事务的执行满足 ACID 原则。

(5)

A dense index does not necessarily improve the efficiency of queries on the indexed attribute.

① If the query involves retrieving all or a significant portion of the records in a small file, using an index may not enhance query performance. Instead, it may introduce additional overhead due to the maintenance and usage of the index.

② If the dense index is a secondary index, not a clustered index, this means that multiple records corresponding to a single key value are scattered across different physical blocks. When there are many records associated with a single key value, retrieving these records can result in increased I/O operations as the database must access multiple physical blocks, which can decrease the efficiency of the query.

稠密索引不一定能提高针对索引属性的查询效率。

① 如果是查询小文件中的全部或相当多的记录时,使用索引并不能提高查询效率,反而会因为索引增加开销;

② 如果稠密索引为次索引，但不是簇集索引，也就是说一个键值对应的多条记录分散在不同的物理块中；当一个键值对应的记录较多时，取这些记录时访问物理块的 I/O 开销反而会降低查询的效率。

2. Try to answer the following questions about concurrency in database systems.

(1) What is concurrency?

The DBMS can accept multiple transactions at the same time, and the transactions can overlap in time.

(2) Why concurrency?

Purpose:

a. Improving system utilization & response time.

b. Different transaction may access to different parts of database.

(3) What problems will concurrency bring?

Problem: Lost update; read dirty data; read value cannot be repeated

(4) What kind of concurrent execution is correct? Why?

Serializable.

The scheduling of concurrent transaction execution being serializable means that the scheduling of these concurrent transactions is equivalent to their serial scheduling. In a serial schedule, the operations of various transactions do not intersect or interfere with each other, thus avoiding conflicts during concurrent execution. Therefore, a concurrent transaction schedule that is equivalent to a serial one will not produce conflicts either, which implies that the outcome is correct.

事务并发执行的调度是可串行化的，也就是说对于该事务并发执行的调度与该事务的串行调度等价；对于串行调度，各事务的操作没有交叉，没有互相干扰，因此不会产生并发执行时的冲突问题，因此与之等价的事务并发调度也不会产生冲突，即并发结果是正确的。

(5) How to avoid problems caused by concurrency?

Concurrency control methods such as locking method and time stamp method can be used

可采用封锁法、时间戳法等并发控制方法

3. What is the basic idea of the locking protocol and how does it ensure that concurrent transactions are executed correctly? What are the problems that must be solved after the adoption of the locking protocol?

Before a concurrent transaction operates on the same data object, it sends a request to the system to lock the operation object. After the transaction's lock request is approved, it has certain control over the object. Before the transaction releases its lock, other transactions cannot obtain the lock request of the data object and operate on it, thus avoiding access conflict and ensuring the correct execution of concurrent transactions.

After adopting the locking protocol, there may be problems such as live lock and deadlock, among which the problem must be solved is the deadlock problem caused by the circular waiting between transactions.

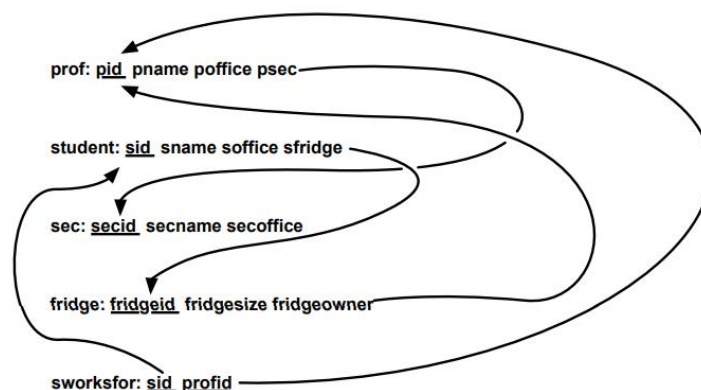
4. Consider a relational table:

```
Professor(
    professor_name, professor_id,
    professor_office_id, student_id,
    student_name, student_office_id,
    student_designated_refrigerator_id, refrigerator_owner_id,
    refrigerator_id, refrigerator_size, secretary_name,
    secretary_id, secretary_office )
```

Suppose the data has the following properties:

- A. Professors and secretaries have individual offices, students share offices.
- B. Students can work for multiple professors.
- C. Refrigerators are owned by one professor.
- D. Professors can own multiple refrigerators.
- E. Students can only use one refrigerator.
- F. The refrigerator the student uses must be owned by one of the professors they work for.
- G. Secretaries can work for multiple professors.
- H. Professors only have a single secretary

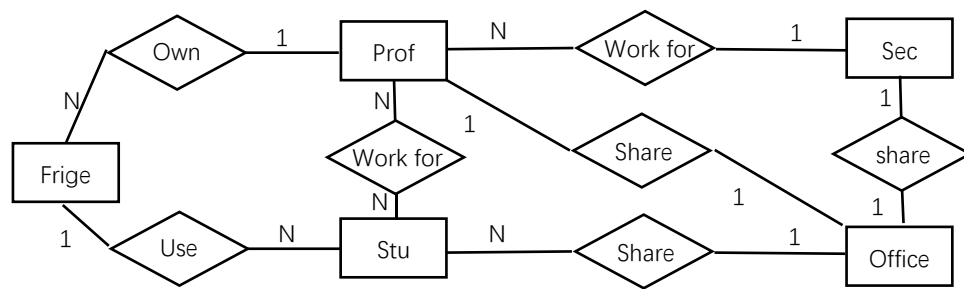
(1) Put this table into 3rd normal form by writing out the decomposed tables; designate keys in your tables by underlining them. Designate foreign keys by drawing an arrow from a foreign key to the primary key it refers to. Note that some of the properties listed above may not be enforced (i.e., guaranteed to be true) by a 3NF decomposition.



(2) Which of the eight properties (A–H) of the data are enforced (i.e., guaranteed to be true) by the 3NF decomposition and primary and foreign keys you gave above.

B,C,D,E,G,H; the schema does not enforce that profs/secretaries have individual offices or that students use the fridge of the professor they work for.

(4) Please draw the E-R diagram of the relationship in this question.



5. Hotel Management System introduction:

One room have one or several beds, and each room has information of room ID (e.g., 201), price and bed number.

Each customer has information of ID, name and gender.

Each reserve record has information of the ID of customer, the ID of room and reserve time.

Each room can be reserved by only one customer at the same time.

Table:

Room(RID, Price, Number);

Customer(CID, Name, Gender);

Reserve(RID, CID, Rdate)

(1) For the relationships in the this question, we have the following query:

SELECT Name

FROM Room, Customer, Reserve

WHERE Room.RID = Reserve.RID AND

Customer.CID = Reserve.CID AND

Room.Price > 500 AND

Customer.Gender = 'female' AND

Reserve.Rdate between [2019.10.01, 2019.10.07]

Please optimize the query and draw the query trees before and after optimization.

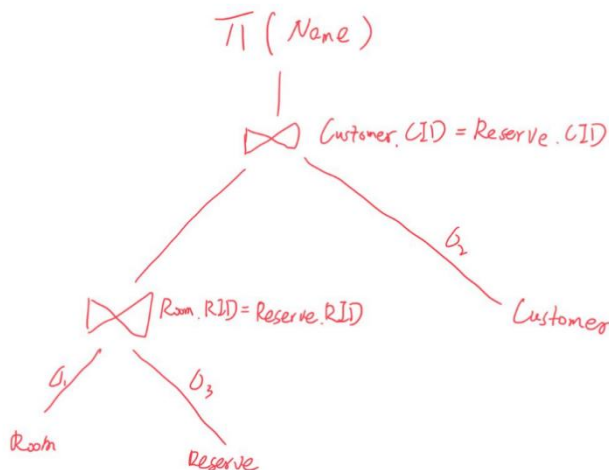
Reference Answer:

σ_1 : Room.Price > 500

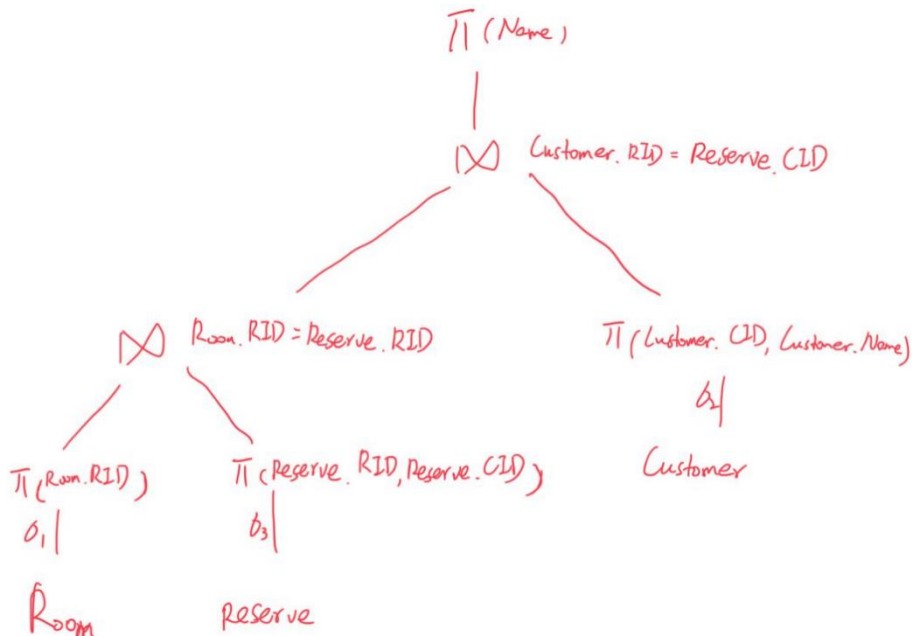
σ_2 : Customer.Gender = 'female'

σ_3 : Reserve.Rdate between [2019. 10. 01, 2019. 10. 07]

Before:



After:



(2) For the Room table in this question, apply Assertion to ensure that the number of rooms ≥ 0 .

Reference Answer:

```
CREATE ASSERTION roomnumber
CHECK ((SELECT COUNT(*) FROM Room) >= 0)
```

6. For the three relational schemas in the above application scenario, write SQL statements that express the following query requirements (must be expressed in a single SQL statement, and WITH is only accepted in question(4)):

- (1) Use join query to find the name of the guest who booked room 210;
- (2) Query the name of the guest who booked all rooms;
- (3) Query the room number and guest name booked by only one person in January 2016;

- (4) For guests who have booked a room more than 60 times, query the maximum number of room bookings per guest in a single month and the year and month corresponding to the maximum number.

Note: The functions for obtaining the year and month of Datetime data in SQL are getYear(...) and getMonth(...) respectively.

(1) Select G.name

From Guest AS G, Accommodation AS A

Where G.ID number=A.ID number And A.room number=210;

(2) Select G.name

From Guest AS G,

Where not exists

(Select R.room number

From Room AS R

Where not exists

(Select A.room number

From Accommodation A

Where A.room number=R.room number

And A.ID number = G.ID number)

);

问题:

- a. 先算出客户预定房间总数（未去重），判断其是否等于房间总数（错误）
客户可能预定一个房间多次

(3) Select A.room number, (Select G.name From Guest G Where G.ID number=A.ID number)

From Accommodation A

Where getYear(A.check-in date)=2016

And getMonth(A.check-in date)=January

Group by A.room number

Having count(distinct G.ID number)=1;

问题:

- a. 对住宿表根据 room number 进行分组后，count(*)=1（错误）
不满足只被一个客户预定
- b. 对住宿表根据 room number, ID number 进行分组，count(*)=1（错误）
结果是对房间只预定了一次

(4) With

GuestTemp(ID number) As

(Select Distinct ID number

From Accommodation

Group by ID number, room number

Having count(*)>60),

AccommodationCount(ID number, year, month, count) As

(Select ID number, getYear(check-in date), getMonth(check-in date),count(*)

From Accommodation

Group by ID number, getYear(check-in date), getMonth(check-in date)),

AccommodationMaxCount(ID number, count) As
 (Select ID number, max(count)
 From AccommodationCount
 Group by ID number),

Select AC.ID number, AC.count, AC.year, AC.month
 From GuestTemp GT, AccommodationCount AC,
 AccommodationMaxCount AMC
 Where AC.ID number=GT.ID number And AC.ID number=AMC.ID number
 And AC.count=AMC.count;

7. The compatibility matrix of (S, X) locks is shown in Figure (a). Suppose that we change the compatibility matrix of (S, X) locks to the one as shown in Figure (b). Now we have n transactions $\{T_1, T_2, \dots, T_n\}$ that require concurrent access in a period of time. Assume that these n transactions are read operations. Will the system work properly? If yes, explain the reason; If not, what happens? (10)

	Locks already owned by other transactions		
Lock request		S	X
	S	Y	N
	X	N	N

(a) Compatibility Matrix A

	Locks already owned by other transactions		
Lock request		S	X
	S	Y	Y
	X	N	N

(b) Compatibility Matrix B

The system will work properly.

Because all of these n transaction are read operations. The read operation only use the s locks. According to the figure(b), when s locks already owned by other transactions, s lock request is compatible. Therefore, n s locks can work at the same time.

8. Please answer the following questions:

(1) Please explain what is Trigger and list the components of Trigger.

Trigger: procedure that starts automatically if specified changes occur to the DBMS

Three parts:

Event (activates the trigger)

Condition (tests whether the triggers should run)

Action (what happens if the trigger runs)

(2) Design a trigger for the SelectCourse table (in the fourth question) to ensure that $0 \leq \text{Grade} \leq 100$ and Sno exists in Student table and Cno exists in Course table when a tuple is inserted.

```

CREATE TRIGGER SelectCourseInsert
AFTER INSERT ON SelectCourse
REFERENCING NEW TABLE as N
FOR EACH STATEMENT
INSERT INTO SelectCourse
    SELECT Sno,Cno,Grade
    FROM N, Student as S, Course as C
    WHERE N.Grade>=0 AND N.Grade<=100 AND N.Sno=S.Sno AND
N.Cno=S.Cno;

```

(3) Try to define a trigger on the accommodation relationship table in question 2 to achieve the following function: For each Insert statement, if the inserted record does not have a check-out date, the operation will be rolled back.

```

Create Trigger insertRollback
Before insert on Accommodation
Referencing New as A
For each statement
When A.check-out date=NULL
Rollback;

```